

1. TASK

Given a photo of a plant leaf, classify it into one of **38 crop-disease classes** (e.g. *Tomato__Late_blight*, *Apple__healthy*). Each class jointly encodes crop type and disease/health status. The output is split into **crop** and **disease** fields for downstream use (advisory recommendations, UI display).

2. DATASET AND EXACT SPLIT

Component	Detail
Base dataset	PlantVillage (Mohanty et al.) — lab-controlled leaf images, uniform background
Supplementary dataset	PlantDoc (Singh et al.) — real-world field/phone-photo leaf images, merged in to reduce lab-to-field domain gap
Total classes	38 (14 crops: Apple, Blueberry, Cherry, Corn, Grape, Orange, Peach, Pepper (bell), Potato, Raspberry, Soybean, Squash, Strawberry, Tomato)
PlantDoc images merged	2,922 images added across overlapping classes
Final training images	45,781
Final test images	11,446
Split ratio	~80% train / ~20% test
Split method	Random shuffle, fixed seed (random.seed(42)), stratification implicit via per-class folder pooling before shuffle

Note on split validity: the test set is a random split of the same merged pool (PlantVillage + PlantDoc), not a fully independent field-only holdout. Reported metrics reflect performance on a distribution similar to training data; true field deployment accuracy may be lower — see Limitations (§13).

3. MODEL / APPROACH

Architecture	ConvNeXt-Tiny (via timm), ImageNet-pretrained backbone, fine-tuned end-to-end
Input	224×224 RGB
Output head	Linear layer, 38-way softmax classification
Baseline compared against	EfficientNet-B0 (previously trained on PlantVillage-only data, no PlantDoc merge)

Why ConvNeXt-Tiny over EfficientNet-B0: ConvNeXt is a modernized CNN architecture that generally outperforms EfficientNet on fine-grained visual classification tasks (small class-discriminating details like lesion texture and color patterns), while remaining lightweight enough (~28M params) for practical training and inference times on free-tier GPUs (Kaggle/Colab T4). It was selected specifically to test whether architecture capacity, combined with better data, would reduce the disease-confusion errors observed with EfficientNet-B0.

4. TRAINING ENVIRONMENT

Platform	Kaggle Notebooks
GPU	Kaggle-provided free-tier GPU (Tesla T4)
Training time	~5.5 min/epoch, 16 epochs run (early-stopped) — ~88 minutes total
Notebook	notebook72125912e4__1_.ipynb
Environment setup	pip install timm datasets huggingface_hub scikit-learn matplotlib seaborn (Kaggle's default Python 3 environment, no additional virtual environment needed)

Training was run entirely on Kaggle's hosted notebook environment rather than locally, to make use of free GPU access and avoid local hardware/driver setup. Dataset (PlantVillage via Hugging Face datasets/zip, PlantDoc via kagglehub) was downloaded directly within the Kaggle session at runtime rather than pre-bundled, and the final checkpoint was uploaded to Hugging Face Hub after training completed, since Kaggle sessions don't persist storage between runs.

5. KEY HYPERPARAMETERS

Parameter	Value
Optimizer	AdamW
Learning rate	5e-5
Weight decay	1e-4
Loss function	Cross-Entropy with per-class weights (inverse class frequency)
Batch precision	Mixed precision (torch.autocast, fp16) with GradScaler
LR scheduler	ReduceLROnPlateau, monitored on validation Macro-F1
Epochs (max)	25
Early stopping patience	5 epochs (no improvement in Val Macro-F1)
Actual epochs run	16 (early-stopped)
Best epoch	11

6. TECHNIQUES USED, RATIONALE FOR THE APPROACH & ALTERNATIVES CONSIDERED

6.1 Class-weighted loss

PlantVillage classes are heavily imbalanced (e.g. Tomato__Tomato_mosaic_virus: 373 images vs Tomato__Late_blight: 1,909 — over 5× difference), biasing predictions toward majority classes. Inverse-frequency weights were applied to CrossEntropyLoss so minority classes contribute proportionally more to the gradient — directly fixing observed mosaic-virus→late-blight misclassifications.

6.2 Domain-gap mitigation via PlantDoc merge

EfficientNet-B0 trained on PlantVillage alone gave confident-but-wrong predictions on real phone-camera images despite 99%+ lab-image validation accuracy — a known issue, since uniform lab backgrounds let models key on lighting/background rather than lesion features. PlantDoc's real backgrounds, varied lighting, and natural leaf angles were merged in to expose the model to field-like conditions.

6.3 Heavier data augmentation

To close the domain gap further without more real data, augmentation was strengthened beyond standard flip/rotate: ColorJitter (lighting), RandomPerspective (camera angle), RandomAffine (framing), and RandomErasing (occlusion) — a standard technique for improving out-of-distribution robustness.

6.4 Early stopping on Macro-F1 (not accuracy or loss)

Macro-F1, not accuracy, drove early stopping and model selection, since accuracy is dominated by large classes (e.g. Orange__Haunglongbing, Soybean__healthy, Tomato__Tomato_Yellow_Leaf_Curl_Virus, each with 1,000+ test samples) and can mask minority-class failures. Macro-F1 weights every class equally, giving an honest signal on whether class-weighting was working.

6.5 Why This Approach Was Chosen

The project went through several iterations, each decision driven by a specific observed failure:

- 1. **Per-crop models (rejected):** confusion-matrix analysis showed near-zero cross-crop confusion — errors were disease-level within a crop, not cross-crop — and several crops (Raspberry, Blueberry, Soybean, Squash) have only one class each, making dedicated models pointless.
- 2. **Root-cause diagnosis:** real-image testing revealed two distinct problems — class imbalance (minority-class confusion) and domain gap (confident-but-wrong predictions despite near-perfect lab accuracy) — not a model-capacity problem.
- 3. **Fix selection:** class weighting was chosen over oversampling/SMOTE (no synthetic data needed, proven for CNN imbalance); PlantDoc merging was chosen over new data collection (public, pre-labeled, overlaps PlantVillage classes) — the fastest path to real-world signal.

6.6 Alternative Approaches Considered

Approach	Why not chosen (for now)
Per-crop dedicated models (9 separate models)	High engineering/maintenance overhead; several crops have only one class; doesn't fix the actual root causes (imbalance, domain gap)
Larger architecture (ConvNeXt-Base/Large, ViT-Large)	Failure mode was data-related, not capacity-related — unlikely to fix OOD errors, while costing more to train
Multi-GPU training	Only cuts wall-clock training time, not accuracy; ~5.5 min/epoch on one free-tier GPU was already practical
Two-stage pipeline (crop classifier → disease classifier, conditioned on crop)	A cheaper inference-time crop-mask variant achieves the same cross-crop-error prevention without a second model or retraining cost

7. EVALUATION TECHNIQUE AND WHY

Standard held-out test set evaluation (11,446 images, disjoint from training via the random split) using scikit-learn's classification report:

- **Accuracy** — overall correct-prediction rate; included because it's the standard baseline-comparable metric, but not used alone due to class imbalance.
- **Macro-F1** — primary metric, unweighted average of per-class F1 scores, chosen specifically because it doesn't let large classes hide small-class failures.
- **Weighted F1** — included for completeness; closer to accuracy since it weights by class support.
- **Per-class Precision/Recall/F1** — used to identify exactly which classes are weak (\$9), not just an aggregate score.
- **Confusion matrix** — computed to identify specific class-pair confusion patterns, not just error rate.

This combination was chosen deliberately: aggregate accuracy alone would have hidden the exact minority-class failure this project needed to diagnose and fix.

8. RESULTS — 80/20 TRAIN/TEST SPLIT

Metric	Value
Accuracy	98.68%
Macro-F1	97.99%
Weighted F1	98.68%
Best Validation Macro-F1 (early stopping checkpoint)	97.99% (epoch 11)

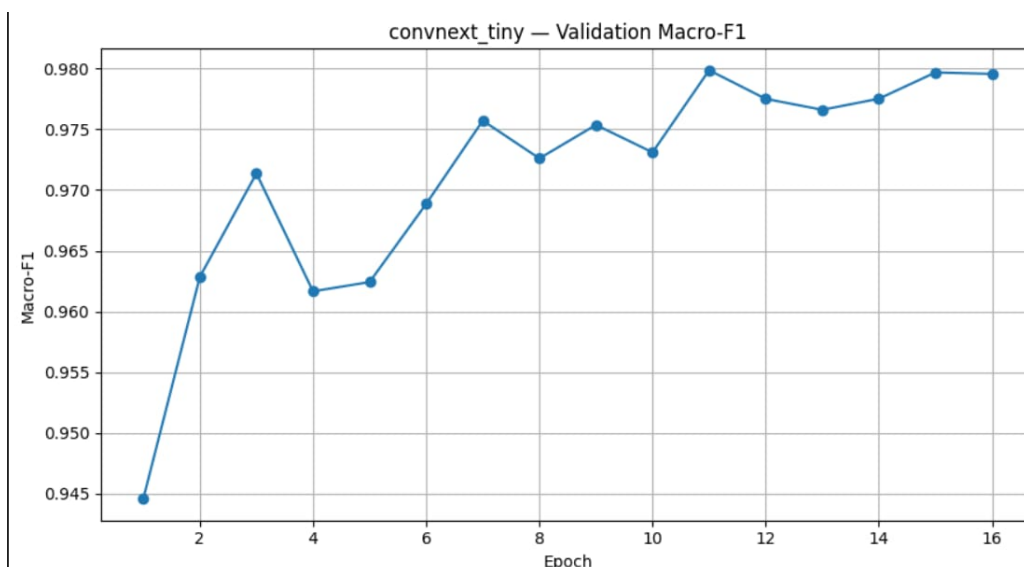


Figure 1. Validation Macro-F1 across training epochs (convnext_tiny).

9. PER-CLASS RESULTS (ALL 38 CLASSES)

Confusion Matrix Graph (All 38 Classes)

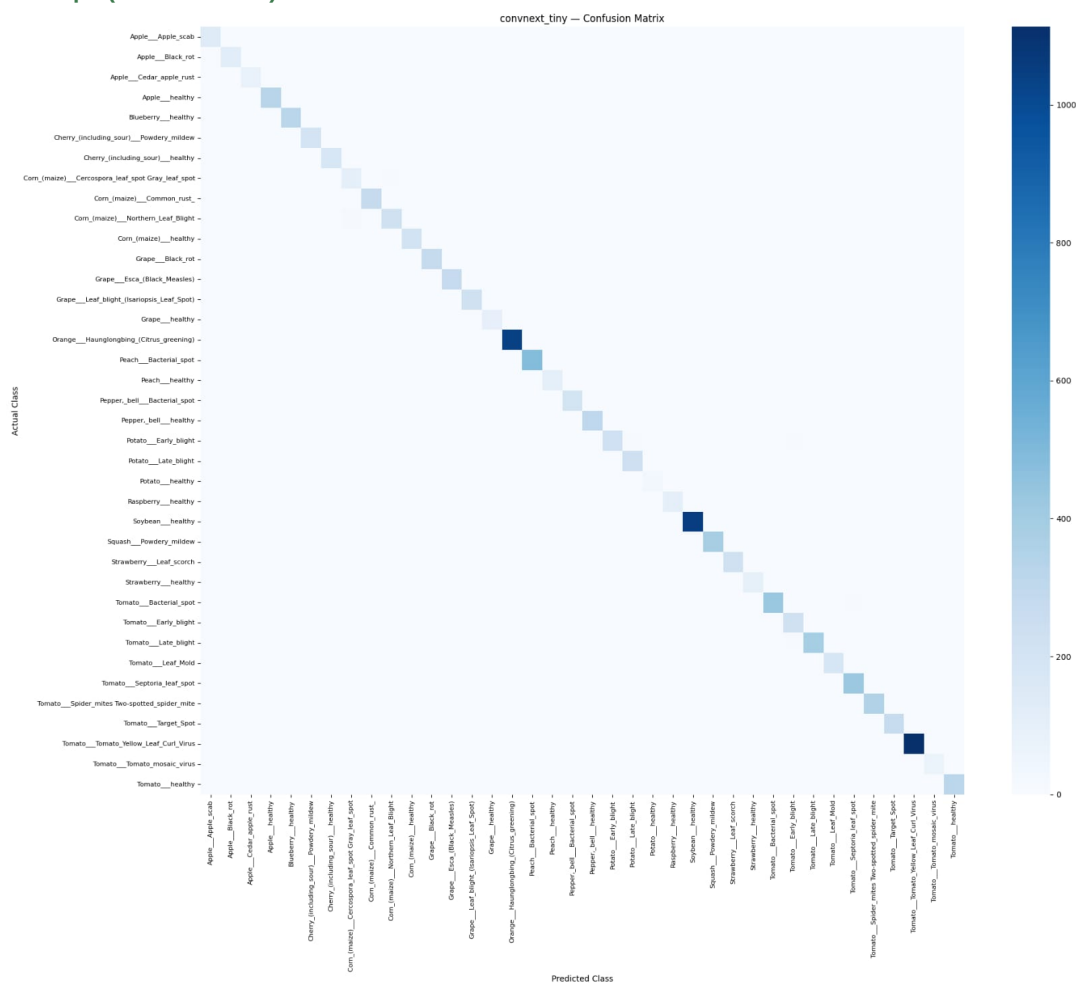


Figure 2. Confusion Matrix Graph (All 38 Classes) — actual vs. predicted class (convnext_tiny, test set).

Class	P	R	F1	N
Apple__Apple_scab	0.9925	0.9301	0.9603	143
Apple__Black_rot	1.0000	1.0000	1.0000	125
Apple__Cedar_apple_rust	0.9425	0.9762	0.9591	84
Apple__healthy	0.9939	0.9939	0.9939	330
Blueberry__healthy	0.9938	0.9846	0.9891	324
Cherry__Powdery_mildew	1.0000	1.0000	1.0000	196
Cherry__healthy	0.9886	0.9775	0.9831	178
Corn__Cercospora_Gray_leaf_spot	0.8843	0.9224	0.9030	116
Corn__Common_rust	0.9858	0.9893	0.9875	280
Corn__Northern_Leaf_Blight	0.9563	0.9241	0.9399	237
Corn__healthy	0.9906	1.0000	0.9953	211
Grape__Black_rot	0.9928	0.9964	0.9946	276
Grape__Esca_(Black_Measles)	1.0000	1.0000	1.0000	281
Grape__Leaf_blight_(Isariopsis)	1.0000	1.0000	1.0000	221
Grape__healthy	1.0000	0.9792	0.9895	96
Orange__Haunglongbing	1.0000	1.0000	1.0000	1039
Peach__Bacterial_spot	1.0000	0.9980	0.9990	488
Peach__healthy	0.9717	1.0000	0.9856	103
Pepper,bell__Bacterial_spot	0.9533	0.9951	0.9737	205

Class	P	R	F1	N
Pepper,bell__healthy	0.9838	1.0000	0.9918	304
Potato__Early_blight	0.9620	0.9344	0.9480	244
Potato__Late_blight	0.9506	0.9665	0.9585	239
Potato__healthy	0.9744	1.0000	0.9870	38
Raspberry__healthy	0.9722	0.9813	0.9767	107
Soybean__healthy	0.9990	0.9990	0.9990	1052
Squash__Powdery_mildew	0.9974	0.9974	0.9974	385
Strawberry__Leaf_scorch	1.0000	1.0000	1.0000	221
Strawberry__healthy	0.9519	1.0000	0.9754	99
Tomato__Bacterial_spot	0.9840	0.9707	0.9773	443
Tomato__Early_blight	0.9325	0.9779	0.9546	226
Tomato__Late_blight	0.9848	0.9726	0.9787	401
Tomato__Leaf_Mold	0.9676	0.9676	0.9676	185
Tomato__Septoria_leaf_spot	0.9792	0.9770	0.9781	434
Tomato__Spider_mites	1.0000	0.9971	0.9986	349
Tomato__Target_Spot	1.0000	0.9926	0.9963	270
Tomato__Yellow_Leaf_Curl_Virus	0.9973	0.9920	0.9946	1122
Tomato__mosaic_virus	0.8846	0.9324	0.9079	74
Tomato__healthy	0.9969	0.9906	0.9937	320

Aggregate	Precision	Recall	F1-Score	Support
accuracy	—	—	0.9868	11446
macro avg	0.9780	0.9820	0.9799	11446
weighted avg	0.9870	0.9868	0.9868	11446

Top classes (F1)	F1	Worst-performing classes (F1)	F1
Strawberry__Leaf_scorch	1.0000	Corn__Cercospora_Gray_leaf_spot	0.9030
Orange__Haunglongbing	1.0000	Tomato__Tomato_mosaic_virus	0.9079
Grape__Leaf_blight_(Isariopsis)	1.0000	Corn__Northern_Leaf_Blight	0.9399
Grape__Esca_(Black_Measles)	1.0000	Potato__Early_blight	0.9480
Cherry__Powdery_mildew	1.0000	Tomato__Early_blight	0.9546

Full per-class precision/recall/F1 source: [scikit-learn classification report](#), [training notebook output \(notebook72125912e4__1_ipynb\)](#).

10. BASELINE COMPARISON

Model	Training data	Val Macro-F1	Notes
EfficientNet-B0	PlantVillage only	~0.99 (on PlantVillage-only test split)	High lab-image accuracy, but confirmed high-confidence misclassifications on real-world test photos (domain gap)
ConvNeXt-Tiny (this model)	PlantVillage + PlantDoc, class-weighted, heavier augmentation	0.9799 (on merged test split, includes harder real-world images)	Slightly lower headline number, but evaluated on a harder, more representative test set including real-world images — not a like-for-like drop in real capability

Important honesty note: the two Macro-F1 numbers are not directly comparable, because the ConvNeXt-Tiny test set includes harder PlantDoc images that the EfficientNet-B0 test set did not. The EfficientNet baseline's near-99% figure reflects an easier, lab-only test distribution. The correct interpretation is that ConvNeXt-Tiny's 97.99% on a harder test set represents genuinely improved real-world robustness, not a regression — but this has only been spot-checked on a handful of manual real-world test images, not systematically benchmarked (see Limitations).

11. KNOWN LIMITATIONS AND FAILURE CASES

- Domain gap is only partially closed.** PlantDoc (2,922 images) is small relative to PlantVillage (54,305 images); real-world accuracy on images very different from both datasets (extreme lighting, multiple leaves, non-leaf backgrounds, blurry phone photos) is not fully validated at scale — only spot-checked manually on a handful of images.
- Test split is not a fully independent field holdout.** The 11,446-image test set is a random split of the same merged pool used for training augmentation source images, not a separate, never-seen real-world benchmark. Reported Macro-F1/Accuracy should be read as in-distribution performance, not a guaranteed field deployment number.
- Weakest classes are visually similar disease variants.** Corn Gray Leaf Spot, Corn Northern Leaf Blight, Potato/Tomato Early Blight are the lowest-F1 classes — these are diseases with genuinely overlapping visual symptoms (necrotic leaf lesions), a known hard sub-problem in plant pathology image classification, not fully resolved by this iteration.
- Minority classes remain relatively weaker despite weighting.** Tomato__Tomato_mosaic_virus (74 test samples) is still the second-worst class — class weighting improved but did not fully equalize performance for the smallest classes.
- No deployment-scale real-world benchmark exists yet.** The confident-wrong prediction issue observed pre-retraining (e.g. Apple scab image predicted as Tomato late blight at 97% confidence) has not been re-verified with a large, systematic real-world test set post-retraining — only individually spot-checked.
- Model outputs a single class with no uncertainty/abstention option.** Low-confidence predictions are not currently flagged or rejected by the model itself; this is handled at the API layer by treating <70% confidence as low-confidence for display purposes only.